

CP1.9: Vectors in 3D

Which paper is this on? Core Pure (Papers 1 and 2) covers only the *scalar* product side of 3D vectors: the vector and Cartesian forms of a line and of a plane (6.1, 6.2), the scalar product and the angles it gives (6.3), checking perpendicularity (6.4), and the intersection of a line and a plane together with perpendicular distances (6.5).

The **vector (cross) product and the scalar triple product are not in Core Pure** — they are section 5 of **Paper 3A: Further Pure Mathematics 1**. They are kept here because they are the quickest route to a normal, an area or a shortest distance, and because most students meet them alongside this topic. Section 3 below, and anything that uses $\mathbf{a} \times \mathbf{b}$, is flagged **FP1**. If you are not sitting Further Pure 1, you can be asked for all of these results by scalar-product methods instead, and you will never be asked for $\mathbf{a} \times \mathbf{b}$ itself.

1. 3D Vector Basics

A 3D vector is written as $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$

Magnitude: $|\mathbf{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$

Unit vector: $\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|}$

Direction Cosines

The direction cosines of a vector $\mathbf{a} = (a_1, a_2, a_3)$ are:

$$\cos \alpha = \frac{a_1}{|\mathbf{a}|}, \quad \cos \beta = \frac{a_2}{|\mathbf{a}|}, \quad \cos \gamma = \frac{a_3}{|\mathbf{a}|}, \quad \text{where } \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

2. Scalar (Dot) Product

Geometric definition:

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta$$

where θ is the angle between the vectors ($0 \leq \theta \leq \pi$).

Component form:

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Key Applications

- **Testing perpendicularity:** $\mathbf{a} \perp \mathbf{b} \iff \mathbf{a} \cdot \mathbf{b} = 0$
- **Finding angle:** $\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|}$
- **Projection of $\mathbf{a}$ onto $\mathbf{b}$:** $\frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|}$

Properties

- $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$ (commutative)
- $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$ (distributive)
- $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$

3. Cross (Vector) Product

FP1 — not Core Pure

Specification reference: Paper 3A Further Pure Mathematics 1, 5.1 (vector product) and 5.2 (scalar triple product).

Core Pure papers assume only the scalar product.

Geometric definition:

$$\mathbf{a} \times \mathbf{b} = |\mathbf{a}||\mathbf{b}| \sin \theta \hat{\mathbf{n}}$$

where θ is the angle between them and $\hat{\mathbf{n}}$ is the unit vector perpendicular to both, following the right-hand rule.

Component form (determinant expansion):

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \mathbf{i}(a_2b_3 - a_3b_2) - \mathbf{j}(a_1b_3 - a_3b_1) + \mathbf{k}(a_1b_2 - a_2b_1)$$

Key Properties of Cross Product

- $\mathbf{a} \times \mathbf{b} = -(\mathbf{b} \times \mathbf{a})$ (anti-commutative)
- $\mathbf{a} \times \mathbf{a} = \mathbf{0}$
- $\mathbf{a} \parallel \mathbf{b} \iff \mathbf{a} \times \mathbf{b} = \mathbf{0}$
- $\mathbf{i} \times \mathbf{j} = \mathbf{k}, \mathbf{j} \times \mathbf{k} = \mathbf{i}, \mathbf{k} \times \mathbf{i} = \mathbf{j}$

Area Applications

$$\text{Area of parallelogram} = |\mathbf{a} \times \mathbf{b}|$$

$$\text{Area of triangle} = \frac{1}{2}|\mathbf{a} \times \mathbf{b}|$$

Example: Find $\mathbf{a} \times \mathbf{b}$ and hence the area of the triangle with vertices A, B, C where $\overrightarrow{AB} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ and $\overrightarrow{AC} = \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix}$.

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & -1 \\ 3 & 0 & 2 \end{vmatrix} = \mathbf{i}(4 - 0) - \mathbf{j}(2 + 3) + \mathbf{k}(0 - 6) = \begin{pmatrix} 4 \\ -5 \\ -6 \end{pmatrix}$$

$$|\mathbf{a} \times \mathbf{b}| = \sqrt{16 + 25 + 36} = \sqrt{77}$$

$$\text{Area of triangle} = \frac{1}{2}\sqrt{77}$$

4. Equation of a Line in 3D

A line through point A with position vector $\mathbf{a}$ and direction vector $\mathbf{b}$ has equation:

Vector form: $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$, where $\lambda \in \mathbb{R}$ is a parameter.

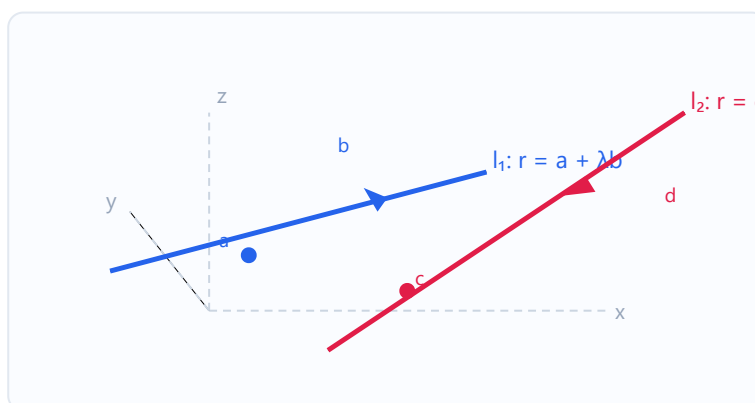
Parametric form:

$$x = a_1 + \lambda b_1, \quad y = a_2 + \lambda b_2, \quad z = a_3 + \lambda b_3$$

Cartesian (symmetric) form:

$$\frac{x - a_1}{b_1} = \frac{y - a_2}{b_2} = \frac{z - a_3}{b_3} = \lambda$$

(provided all components of $\mathbf{b}$ are non-zero)



Example: Write the line through $A = (2, -1, 3)$ and $B = (5, 1, -2)$ in all three forms.

Direction vector: $\overrightarrow{AB} = \begin{pmatrix} 3 \\ 2 \\ -5 \end{pmatrix}$

Vector form: $\mathbf{r} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 2 \\ -5 \end{pmatrix}$

Parametric: $x = 2 + 3\lambda$, $y = -1 + 2\lambda$, $z = 3 - 5\lambda$

Cartesian: $\frac{x-2}{3} = \frac{y+1}{2} = \frac{z-3}{-5}$

5. Equation of a Plane

A plane with normal vector $\mathbf{n} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ passing through point A with position vector $\mathbf{a}$:

Scalar product form: $\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n} = d$

Cartesian form: $ax + by + cz = d$

Parametric form: $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c}$ where $\mathbf{b}$ and $\mathbf{c}$ are two non-parallel direction vectors in the plane.

Example: Find the equation of the plane through $A(1, 0, 2)$, $B(3, 1, 0)$, $C(0, 2, 1)$.

Find two direction vectors: $\overrightarrow{AB} = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$, $\overrightarrow{AC} = \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$

Normal: $\mathbf{n} = \overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & -2 \\ -1 & 2 & -1 \end{vmatrix}$

$$= \mathbf{i}(-1 + 4) - \mathbf{j}(-2 - 2) + \mathbf{k}(4 + 1) = \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}$$

$$d = \mathbf{a} \cdot \mathbf{n} = 1(3) + 0(4) + 2(5) = 13$$

$$\text{Plane: } 3x + 4y + 5z = 13$$

6. Angles Between Lines and Planes

Angle between	Formula	Notes
Two lines (direction $\mathbf{b}_1, \mathbf{b}_2$)	$\cos \theta = \frac{ \mathbf{b}_1 \cdot \mathbf{b}_2 }{ \mathbf{b}_1 \mathbf{b}_2 }$	Use $ \cdot $ to get acute angle
Two planes (normals $\mathbf{n}_1, \mathbf{n}_2$)	$\cos \theta = \frac{ \mathbf{n}_1 \cdot \mathbf{n}_2 }{ \mathbf{n}_1 \mathbf{n}_2 }$	Use $ \cdot $ to get acute angle
Line (direction $\mathbf{b}$) and plane (normal $\mathbf{n}$)	$\sin \theta = \frac{ \mathbf{b} \cdot \mathbf{n} }{ \mathbf{b} \mathbf{n} }$	Note: $\sin$, not $\cos$

Exam tip: For the angle between a line and a plane, the formula uses $\sin$ (because the line makes an angle θ with the plane but $90^\circ - \theta$ with the normal). This is the most common error in this section.

7. Intersections

Line–Line Intersection (or Skew)

For two lines $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$ and $\mathbf{r} = \mathbf{c} + \mu\mathbf{d}$:

1. Set the x, y, z components equal to get three equations in λ and μ
2. Solve two of the equations for λ and μ
3. Check if these values satisfy the third equation
4. If yes: lines intersect (substitute to find point). If no: lines are skew or parallel

Parallel lines: Direction vectors are parallel ($\mathbf{b} = k\mathbf{d}$ for some scalar k).

Skew lines: Not parallel and do not intersect.

Line–Plane Intersection

Substitute the parametric equations of the line into the plane equation to find λ , then find the point.

Example: Find where $\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$ meets the plane $x - 2y + z = 5$.

Substitute parametric form: $(1 + 2\lambda) - 2(2 + \lambda) + (-1 + 3\lambda) = 5$

$$1 + 2\lambda - 4 - 2\lambda - 1 + 3\lambda = 5$$

$$3\lambda - 4 = 5 \Rightarrow \lambda = 3$$

Point: $x = 7, y = 5, z = 8$. Intersection at $(7, 5, 8)$.

Plane–Plane Intersection

Two non-parallel planes intersect in a line. Find the direction vector using the cross product of the two normals. Then find a specific point on the intersection (set one variable to 0 and solve the system of two equations).

8. Distance Formulae

Distance from point P to plane $\mathbf{r} \cdot \mathbf{n} = d$:

$$\text{Distance} = \frac{|\mathbf{p} \cdot \mathbf{n} - d|}{|\mathbf{n}|}$$

where $\mathbf{p}$ is the position vector of P .

In Cartesian form (plane $ax + by + cz = d$, point (x_0, y_0, z_0)):

$$\text{Distance} = \frac{|ax_0 + by_0 + cz_0 - d|}{\sqrt{a^2 + b^2 + c^2}}$$

Distance from point A to line $\mathbf{r} = \mathbf{b} + \lambda \mathbf{d}$:

$$\text{Distance} = \frac{|\overrightarrow{BA} \times \mathbf{d}|}{|\mathbf{d}|}$$

where B is any point on the line.

Shortest distance between two skew lines $\mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$ and $\mathbf{r} = \mathbf{c} + \mu \mathbf{d}$:

$$\text{Distance} = \frac{|(\mathbf{c} - \mathbf{a}) \cdot (\mathbf{b} \times \mathbf{d})|}{|\mathbf{b} \times \mathbf{d}|}$$

Example: Find the shortest distance between the skew lines

$$l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \text{ and } l_2 : \mathbf{r} = \begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$$

$$\mathbf{b} \times \mathbf{d} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \cdot 2 - 0 \cdot (-1) \\ 0 \cdot 1 - 2 \cdot 2 \\ 2 \cdot (-1) - 1 \cdot 1 \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \\ -3 \end{pmatrix}$$

$$|\mathbf{b} \times \mathbf{d}| = \sqrt{4 + 16 + 9} = \sqrt{29}$$

$$\mathbf{c} - \mathbf{a} = \begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix}$$

$$(\mathbf{c} - \mathbf{a}) \cdot (\mathbf{b} \times \mathbf{d}) = -2 - 8 - 12 = -22$$

$$\text{Shortest distance: } \frac{|-22|}{\sqrt{29}} = \frac{22}{\sqrt{29}} = \frac{22\sqrt{29}}{29}$$

9. Summary of Key Vector Formulae

Result	Formula
Dot product	$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3 = \mathbf{a} \mathbf{b} \cos \theta$
Cross product magnitude (FP1)	$ \mathbf{a} \times \mathbf{b} = \mathbf{a} \mathbf{b} \sin \theta$
Area of triangle (FP1)	$\frac{1}{2} \mathbf{a} \times \mathbf{b} $
Line equation	$\mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$
Plane equation	$\mathbf{r} \cdot \mathbf{n} = d$ or $ax + by + cz = d$
Angle between lines	$\cos \theta = \frac{ \mathbf{b}_1 \cdot \mathbf{b}_2 }{ \mathbf{b}_1 \mathbf{b}_2 }$
Angle between planes	$\cos \theta = \frac{ \mathbf{n}_1 \cdot \mathbf{n}_2 }{ \mathbf{n}_1 \mathbf{n}_2 }$
Angle: line and plane	$\sin \theta = \frac{ \mathbf{b} \cdot \mathbf{n} }{ \mathbf{b} \mathbf{n} }$
Point to plane distance	$\frac{ \mathbf{p} \cdot \mathbf{n} - d }{ \mathbf{n} }$
Point to line distance (FP1 form)	$\frac{ \overrightarrow{BA} \times \mathbf{d} }{ \mathbf{d} }$
Skew lines distance (FP1 form)	$\frac{ (\mathbf{c} - \mathbf{a}) \cdot (\mathbf{b} \times \mathbf{d}) }{ \mathbf{b} \times \mathbf{d} }$

Common exam mistakes in vectors:

- **Line–plane angle:** Must use $\sin \theta$ with the NORMAL, not $\cos \theta$.
- **Skew vs parallel:** Always check both conditions. Skew means not parallel AND not intersecting.
- **Normal to plane from parametric form:** Take the cross product of the two direction vectors.
- **Shortest distance:** Always verify the lines are actually skew before applying the formula.
- **Exact answers:** Leave distances in surd form unless told to give a decimal.